

JIAHAO (BRYSON) ZHU

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CORE SKILLS

- **Programming:** C / R / Java / Python / Lisp **Language:** English (Fluent) / Chinese (Native)
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PROJECTS

CO detection & warning device by Raspberry Pi 3 <https://youtu.be/rL4GJKw2br0> 2024.3 / Python

This project's core idea was to create a system that can measure the level of CO and detect gas leaks by using MQ-2. I connect the MQ-2 to the MCP3008 and Pi on breadboard with several wires, import GPIO libraries and define the pins of Pi. MCP3008 uses the SPI for communication with the Pi and convert the analog signals of CO level to digital signals. After calculation, there are two speakers will be played.

Gantt Chart Generator 2025.3 / C

The Gantt Chart is displayed as a formatted table with task names, corresponding months and a column for the dependencies of each task. My work is the code of circular dependencies. The process of searching for circular dependencies in the Gantt Chart is implemented using Depth-First-Search (DFS) algorithm to systematically traverse the task dependency array. It starts from a node(Task 1) and traverse all tasks to determine if there are any circular dependencies.

UCD Swap <https://github.com/Zahotoo/UCDSWAP> 2025.11 – present / Java

I developed a Spring Boot backend for a university-based trading platform, implementing a complete user registration flow with password encryption, email verification, and access control. The system uses token-based email verification with expiration handling to ensure account security, and persists user data in a MySQL database via JPA/Hibernate. I configured Spring Security to protect application endpoints and tested all APIs using Postman, with email functionality verified through Mailtrap. Through this project, I learned how to design and debug a real authentication workflow rather than relying on framework defaults.

EDUCATION

BSc Computer Science | University College Dublin 2024 - 2028 / Dublin

- **Stage 1 GPA:** [3.55/4.2]
 - **Key modules:** Computer Programming (C) / Functional Programming / Statistics with Python
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OTHER ACTIVITIES

- Self-studying Amazon Web Service (AWS) Practitioner.
- Participated in the NASA International Space Apps Challenge 2025 and developed a weather app.
- Working as a cashier & sales at MUJI from 2023.5 – 2023.7 in Shanghai.
- Enjoy playing soccer, hiking, taking photos and video games. Has 140K views of my photos.
- Evaluate the power law exponent (PLE) based on item and Gini Index for over 20 well-known datasets of the recommender system under the "Recbole" framework, which helps compare the long-tail effect and popularity bias of the datasets.